Education and Information Technologies
https://doi.org/10.1007/s10639-024-13043-w

Integrating unplugged and plugged activities for holistic AI education: An embodied constructionist pedagogical approach

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Received: 16 January 2024 / Accepted: 5 September 2024
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Abstract
There is a growing consensus that AI literacy requires a holistic lens, including not only technical knowledge and skills but also social and ethical considerations. Yet, providing holistic AI education for upper-primary students remains challenging due to the abstract and complex nature of AI and a lack of pedagogical experiences in schools. Against this backdrop, this study employs a design-based research (DBR) methodology to develop pedagogical solutions and theoretical knowledge. Drawing upon theories about AI literacy and learning sciences, this study proposes a novel pedagogical approach, namely, the embodied constructionist approach. This approach, by integrating embodied analogies and constructionist making, forges a cyclical *understand–make–reflect* process. Through this process, students gradually comprehend the conceptual, technical, and ethical dimensions of AI, towards holistic development in AI literacy. This approach was implemented and fine-tuned in authentic classrooms with 107 sixth graders over two years. During the implementation, a variety of qualitative data were collected from classroom observation, video-recording, student interviews and artifacts, and debriefings with the teacher. Using a case studies method, this study uncovered the overall learning progress demonstrated by the class in AI knowledge, skills, and ethical and critical reflections, as well as the reflections and struggles experienced by individual students. This study offers an age-appropriate pedagogy to promote inclusive, holistic AI education in the crucial, yet under-examined upper primary context. It highlights the potential of integrating AI literacy and computational thinking (CT) in K-12 education, while contributing to theoretical knowledge about embodied cognition and constructionism in the context of AI education.

Keywords AI literacy · AI education · Pedagogy · Embodied cognition · Constructionism · Computational thinking

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Published online: 11 October 2024

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1 Introduction

Rapid advancements in artificial intelligence (AI) technologies and their impact on modern society, especially the recent breakthrough in generative AI models, have triggered a growing interest in equipping today's schoolchildren with AI literacy (Kreinsen & Schulz, 2023; UNESCO, 2022). As a newly emerging competence, AI literacy represents a fundamental skillsets and understandings necessary to strive in an AI-infused world (Long & Magerko, 2020). This includes not only the technical aspects of how AI systems function and are developed, but also an awareness of the ethical, social, and cultural implications of AI in society (Druga et al., 2019; Touretzky et al., 2019). Despite its significance, AI literacy education remains underexamined in upper primary schools, the stage at which AI literacy is often introduced in K–12 education (UNESCO, 2022). The gap between the complexity of AI and young learners' cognitive abilities makes pedagogical design a significant challenge (Wong et al., 2020).

A key issue is the lack of age-appropriate pedagogies (Casal-Otero et al., 2023). A recent systematic review reveals very limited teaching cases situated in the upper primary context and most of the existing cases involve playing with robots and training machine learning (ML) models on specialized platforms (Ng et al., 2023). These activities, though experiential and engaging, tend to yield only a superficial understanding and application of AI. This is primarily because students find it challenging to link these isolated tasks with broader AI concepts (Zimmerman, 2018). Meanwhile, AI literacy implies a holistic approach of integrating conceptual, technical, and ethical dimensions in students' AI learning, which adds an extra layer of complexity to the pedagogical design (Gardner-McCune et al., 2019). Recognizing this, the challenge becomes how to support holistic AI learning in upper primary AI education in an age-appropriate manner.

Against this research background, this design-based research (DBR) aims to design a novel pedagogical approach that supports holistic AI literacy development among upper-primary students. Specifically, this DBR study will develop a theory-informed pedagogical approach as a practical solution, and then conduct empirical research to advance theoretical insights into this approach (Anderson & Shattuck, 2012). Following the DBR convention, we start with a literature review to delineate existing pedagogical challenges and synthesize relevant theories to inform the pedagogical design. Informed by the theories, we propose an embodied constructionist approach as a pedagogical solution and implement it in local schools for empirical investigation. During the implementation, we will examine the perspective and experience of participating students to draw evidence-based conclusions about the impact of this approach. We will further discuss the implications of research findings for AI education research and practices.

2 Literature review

2.1 A holistic approach to AI literacy

AI literacy refers to the essential competencies that people need to live, learn, and work with AI technologies, to communicate and collaborate effectively with AI, and to

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critically evaluate AI and its impacts (Long & Magerko, 2020). Through an exploratory review, Ng et al. (2021) synthesize four aspects of fostering AI literacy: knowing and understanding AI, using and applying AI, evaluating and creating AI, and AI ethics. Meanwhile, UNESCO (2022), in a global review of K-12 AI curricula, categorizes AI learning objectives into three dimensions:

(i) **Knowledge**: focused on students' cognitive development, including understanding and comprehending basic AI concepts, processes, principles, and applications.
(ii) **Skills**: focused on students' psychomotor development, including the skills, techniques, and strategies needed to apply, design, and create AI systems in various contexts.
(iii) **Values and attitudes**: focused on students' social and emotional development, including the values and attitudes needed to evaluate and respond to ethical issues responsibly.

Multifaceted AI literacy implies a holistic approach to AI education, requiring comprehensive considerations during the pedagogical design (Casal-Otero et al., 2023).

To promote holistic AI literacy, Long and Magerko (2020) suggest integrating *plugged* and *unplugged* activities. Plugged activities employ technologies, such as digital and mobile applications and block-based programming, for hands-on engagement and problem-solving exercises (Ng et al., 2023; Yang, 2022). By contrast, unplugged activities incorporate minimal to no technologies, such as puzzles and physical games, and usually focus on AI concepts and thinking skills (Lindner et al., 2019). Currently, plugged activities are more prevalent in upper primary schools and, if integrated, often appear together with direct instruction (Yue et al., 2022). This kind of integration, though common in postsecondary computer science education, appears to be less ineffective in primary schools, as direct instruction on AI concepts is too abstract for young students still developing their formal operational thinking (Olari et al., 2021; Piaget, 2013). As a result, many students struggle to grasp AI concepts and begin replicating teachers' demonstrations during hands-on activities, without truly understanding the underlying principles (Zimmerman, 2018). As such, one question that remains is how to integrate the conceptual, technical, and ethical dimensions of AI education so they are age-appropriate for upper primary students.

2.2 Developing AI skills through the constructionist pedagogy

Plugged, hands-on engagement with technology, considered a cornerstone of AI education, resonates with constructionist learning and the maker movement in STEM education. Constructionism posits that children learn most effectively when actively creating tangible objects, which can be physical or digital, in the real world (Papert, 1991). Central to this theory is the idea of "learning by making," where the creation of external, tangible artifacts serves as a mediator and representation of learning (Ackermann, 2001; Ng, 2021). Kahn and Winters (2021) trace the historical origin of the constructionist AI pedagogy to the 1970s, when Papert and Solomon (1971) assigned such AI projects as basic natural language processing, into Logo games to teach computational concepts.

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Nowadays, constructionist projects focused on creating AI models and systems have been widely adopted in AI education. This pedagogy has been found effective in forging active participation, as well as programming, problem-solving, and other essential skills among students (Ali et al., 2019; Sysoev et al., 2022). Constructionist approaches also echo computational thinking (CT), a prominent topic in contemporary STEM education, due to their shared emphasis on problem-solving and hands-on engagement (Weng et al., 2024). As such, the growing adoption of CT in K-12 schools may provide a strong foundation that can facilitate the introduction and implementation of constructionist AI education programs.

A critical challenge of the constructionist pedagogy is that creating AI systems firsthand is too difficult for upper primary students (Olari et al., 2021). This issue has been addressed by recent advances in AI teaching tools, such as *Machine Learning for Kids* (Lane, 2021), *RAISE AI Playground* by the MIT Media Lab (2021), and *Teachable Machine* by Google (2017). These tools, often embedded into block-based programming platforms like Scratch, allow students to design and develop their own AI models through intuitive actions, including clicking, dragging, and dropping (Weintrop, 2019). With these child-friendly tools, young students without a technical background can create AI-powered systems that can sense, converse, classify, reason, and move (Williams, 2022). Moreover, age-appropriate strategies have been developed to support students' constructionist engagement. One strategy is embodied experimentation, where students use their bodily actions (e.g., hand/body gestures) as data inputs to train ML models and develop AI-powered systems (e.g., gesture recognition systems; Shamir & Levin, 2022; Williams et al., 2019). In this way, students' bodily actions become a technical resource for AI system development, making the tech-centric projects personally relevant and meaningful to students.

Nevertheless, constructionist pedagogy often faces criticism for its overemphasis on technical skills and operation at the expense of deep and critical learning (Druga et al., 2019; Ruppert et al., 2023). In many constructionist AI curricula, ethical and critical reflections appear as mere "add-ons" to hands-on projects, rather than as integral components of the foundational learning process (Reich et al., 2020). This "add-on" approach tends to place critical reflections as secondary to technical proficiency, sidelining the importance of AI ethics (Lee et al., 2022). Thus, future research is needed to explore how the constructionist pedagogy can be restructured to adequately accommodate the ethical dimension for holistic AI education.

2.3 Teaching AI concepts and ethics with embodied cognition

While the plugged constructionist pedagogy can aid in training AI skills, educators have increasingly adopted the unplugged approach, using *embodied* activities to teach AI concepts (Lindner et al., 2019). This strategy often incorporates physical games and activities, through which students use their bodily movements and sensations to grasp concepts in a tangible, interactive way. For example, in teaching machine learning (ML), students imagine themselves as learning machines: as they "learn" or gather more training data (represented by objects like blocks or cards), their game "performance" improves, thus simulating the ML process. By eschewing technology, this embodied strategy can alleviate the anxiety and fears often associated with unfamiliar technology, rendering AI

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concepts accessible to young students in an age-appropriate way (Heinze et al., 2010; Long & Magerko, 2020).

Underpinning this strategy is the theory of embodied cognition, which emphasizes the crucial role of the body in cognitive processing (Shapiro, 2019). Rather than viewing the mind as an isolated entity, this theory posits that sensory perceptions, physical actions, and gestures are integral to the learning process (Kiverstein, 2012). This theory has been increasingly adopted in STEM education because it bridges unfamiliar ideas with familiar sensorimotor experiences, often via conceptual metaphors or analogical mapping (Weisberg & Newcombe, 2017). Drawing on this, Druga et al. (2019) suggest embodied simulation, where students enact AI processes as robots or intelligent agents, akin to the above ML activity. While this strategy offers a tangible and interactive approach to introducing AI concepts, emerging evidence suggests that such activities might be too complex for young students who are new to the subject matter (Beier & Rau, 2022; Wilensky et al., 1999). For example, students may struggle with thinking/acting like AI but defaulted to their innate human logic. This struggle was largely caused by their unfamiliarity with AI and the challenges of perspective-taking and understanding the underlying conceptual metaphor, which potentially produced a cognitive overload (Ke et al., 2019).

To address these challenges, Dai et al. (2023a) proposed a domain-specific strategy, termed the *embodied analogy*, leveraging human experiences to teach AI, in contrast to simulations. It aligns with a prevailing belief within the AI community that understanding AI requires defining intelligence first (Russell & Norvig, 2021). Historically and theoretically, humans have served as a key reference point for conceptualizing intelligence in AI. In line with this perspective, an embodied analogy usually begins with unplugged physical games, steering students to observe and reflect on human interactions with their environments, such as through obstacle avoidance or object recognition. These human experiences then serve as a basis to explain AI processes by juxtaposing them with human capabilities. In this way, embodied analogy provides a relatable, concrete basis for learning AI, without relying on abstract metaphors. Beyond AI concepts, embodied analogy is also helpful in triggering students' critical reflections, as human–AI comparisons provide a tangible framework for students to assess AI strengths and weaknesses (Dai et al., 2024).

The positive effects of the embodied analogy on AI knowledge and ethics suggest its potential to complement the constructionist pedagogy which focuses on AI skills. While the embodied analogy is primarily an unplugged approach, integrating it with a plugged constructionist pedagogy may fill the research gap present in holistic AI literacy education. Yet, integration involves more than simply combining the two; a thoughtful synthesis aimed at crafting a seamless and mutually enriched learning journey is necessary. Moreover, the integration should be empirically tested and validated in authentic classrooms.

3 Theory-informed design: An embodied constructionist approach

Guided by the literature, we propose an embodied constructionist approach for holistic AI literacy in upper primary education. This approach, illustrated in Fig. 1, is designed as a cyclical learning process with interconnected activities that reinforce each other.

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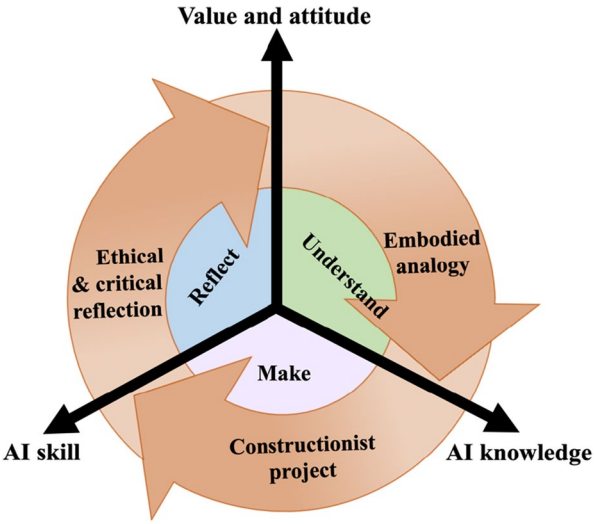


Fig. 1 The structure of the embodied constructionist approach to developing AI literacy

This approach includes the following components:

(i) **An embodied analogy** is used to teach AI concepts and processes through bodily engagement and comparative mapping. This strategy uses students' bodily experiences of performing intelligent tasks as an anchor, and the comparative mapping between humans and AI in performing a similar task aids in explaining AI attributes, processes, and strategies.

(ii) **A constructionist project**, following the embodied analogy, is introduced to develop AI-powered systems and practice students' programming and problem-solving skills. In these projects, while actualizing AI processes and systems with programming tools, students gain first-hand insights into how AI processes differ from human processes.

(iii) **Ethical and critical reflections** are woven into embodied, constructionist activities. Students explore AI's strengths and limitations through ongoing human–AI comparisons, and they evaluate the ethical implications of their digital creations critically.

Through this learning cycle, students can consistently draw parallels between humans and AI to understand, apply, and reflect on AI concepts and practices (Bloom & Krathwohl, 2020). By integrating unplugged and plugged activities, this approach introduces an understand–make–reflect cycle that combines the conceptual, technical, and ethical dimensions of AI. In this way, the approach is expected to forge the holistic development of students' AI literacy through intuitive, relatable, and tangible learning experiences. Figure 2 demonstrates this approach through two lesson plans on intelligent agents and ML.

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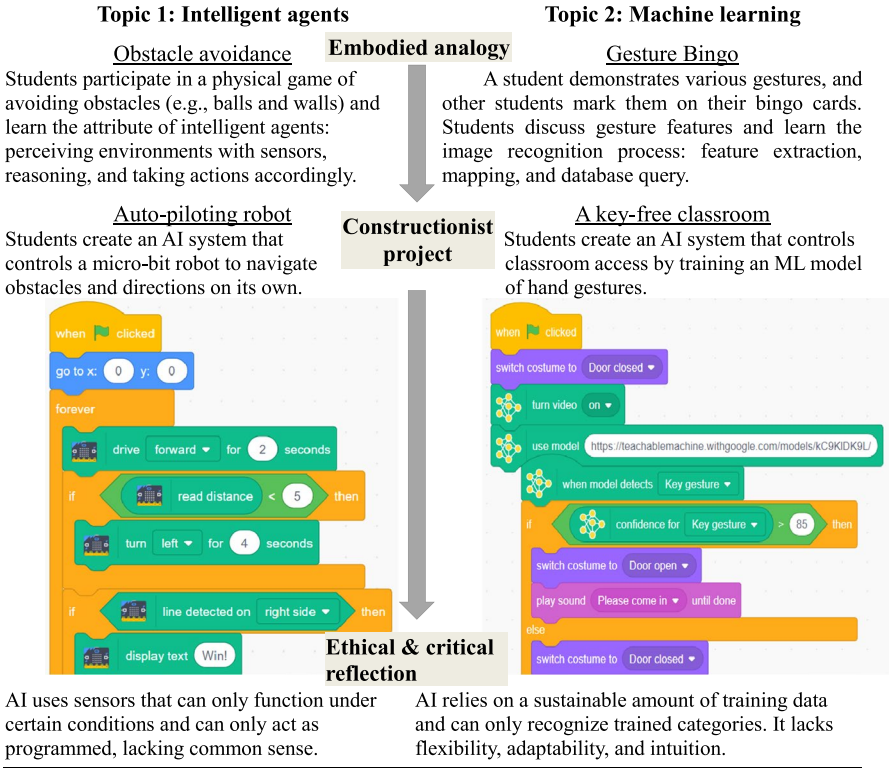


Fig. 2 Examples of lesson plans informed by the embodied constructionist approach

4 Empirical investigations and research methods

Following the DBR convention, the embodied constructionist approach was developed and refined through two iterative cycles of (re)design, enactment, and analysis. During the enactment in classrooms, we conducted empirical investigations to investigate its actual impacts on students' AI literacy development. The investigation was guided by the following research questions (RQs):

> *How does the embodied constructionist approach impact upper primary students' AI literacy in comparison with their pre-instructional conceptions?*

To address the RQ, we conducted a case study in the AI summer camp in the main study. The case study allows for an in-depth examination of the novel pedagogy in practice, capturing the complexities and nuances of real-world application within the classroom setting (Hatch, 2002; Yin, 2018).

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4.1 Research context and participants

This study was situated in an ongoing school-university partnership project where researchers and schoolteachers collaborated to introduce AI literacy to primary schools (Dai et al., 2020; Dai et al., 2023b). Part of the partnership project was to co-develop an AI summer camp for students, which served as the context for implementing the embodied constructionist approach. The first DBR cycle in 2022, involving 75 students, served as the pilot study, and it focused on prototype development and collection of feedback for improvement. The second cycle in 2023 – the main study – involved 32 students.

The partner school is a public school situated in a low socio-economic community in Hong Kong (HK). Most students at this school are from a single-income household. The school has an average sociocultural background, where most students and teachers are Han ethnicity. Two months before the summer camp, a public letter was sent to all students and parents in the fifth graders to recruit participants in the summer camp. From the applicants, 32 students were selected with these criteria: 1) no previous learning experiences with AI or relevant subjects like robotics; 2) commitment to have full attendance of the summer camp; 3) a gender ratio of 1:1; 4) diversity in academic stances. The average age of the 32 students was 11 years old. Informed consent was gained from the students, parents, teachers, and school principal before the study.

The summer camp comprised eight 3-h workshops in the morning of weekdays. The teaching content, as shown in the Appendix Table 2. The teaching content was informed by the curriculum framework "Five Big Ideas of AI" (Touretzky et al., 2019) and the K–12 AI curriculum report by UNESCO (2022). The content focused on basic topics of AI, from introductory computational concepts (e.g., binary number and features) to machine learning (e.g., supervised learning and recommender systems), where AI programming and ethical issues were integrated. In alignment with the embodied constructionist approach, the first half of the 3-h workshop took place in a regular classroom with movable tables and chairs – a space ideal for conceptual teaching and embodied activities; the second half took place in a computer lab and each student was equipped with a computer for constructionist projects. In HK, Technology Education in the upper primary context generally focuses on developing computational thinking via the block-based programming platform Scratch, developed by the MIT Media Lab (EDB, 2020). Given this context, the platform *RAISE AI Playground* (Lab, 2021), which is built upon Scratch but with additional AI-specific blocks, was adopted for constructionist projects in this study.

4.2 Data collection

A series of research activities took place to collect data, as summarized in Table 1. Before the study, a drawing assessment was administered with the teacher's help to establish the baseline of learning. Drawing allows young students to externalize their thoughts of abstract concepts with visual representations (Chang et al., 2020). Students were tasked to use visual elements and texts to show what they thought AI was within 15 min. They were also allowed to skip this task by indicating they had no idea.

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Table 1 A summary of data collection

Source	Content	Volume
Student	Video of whole-class interactions	3 h × 8 workshops = 24 h
	Drawing pieces before the study	1 piece × 30 students = 30 sets
	Learning artefact (worksheets and Scratch projects)	8 workshops × 32 students = 256 sets
	Individual interview of case students	20 min × 9 rounds of interviews × 8–10 students ≈ 25 h
Teacher	Post-teaching debriefing with researchers	30 min × 8 workshops ≈ 4 h
Researcher	Observation fieldnote	8 sets

During the workshops, researchers attended as teaching assistants and conducted participant observation (Spradley, 2016). Through the participant observation, the researchers gained first-hand insights into classroom dynamics and student engagement and built rapport with the students. Particularly, the researchers would circulate the classroom/lab during the constructionist projects or any group activities to offer scaffolding when necessary, while observing the learning processes and problem-solving strategies of students. The researchers also documented their observations and reflections as fieldnotes, capturing both the explicit activities and the tacit nuances of the classroom environment, such as non-verbal cues, emotional responses, and levels of student participation. Meanwhile, the classroom dynamics were video-recorded and student artifacts (e.g., worksheets and Scratch projects) were collected. After each workshop, the researchers and teacher had a post-teaching briefing (approximately 30 min) to review student participation and reflect on their classroom delivery. Such simultaneous discussion allowed researchers to collect feedback and insights from teachers and clarify any ambiguous or confusing observations, enhancing the overall quality and effectiveness of data collection (Vosniadou & Kampylis, 2013). The debriefing was audio-recorded and transcribed verbatim.

After the *drawing assessment* and *each workshop* of the summer camp, eight to ten students were invited for a stimulus-recall interview individually, each lasting for approximately 20 min (Dempsey, 2010). The interviewee students were selected according to their performance and participation in the respective assignments. The selection mainly considered representativeness and variation to fully understand typical student experiences and individual differences (Yin, 2018); factors like gender, baseline understanding, and academic performance were considered. In the interview, collected artifacts (e.g., the student drawing, worksheet, or Scratch project) were displayed as a stimulus to encourage the student to recall his/her thinking process and thoughts during these activities. The interview followed a semi-structured format, with pre-set questions covering that day's focal activities and content, but the interviewer can add more questions based on the flow of conversation. The interview focused on students' cognitive process in the embodied activities and constructionist projects (e.g., their block selection, problem-solving methods, and connections with AI concepts and ethical issues). All interviews were video-recorded.

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4.3 Data analysis

Considering the goal of understanding the actual impacts of the proposed approach, multiple qualitative methods were combined to probe the authentic classroom implementation (Dorman, 2002; Hopkins, 2014). In alignment with the holistic view of AI literacy, the analysis focused on three dimensions: AI knowledge, skills, and values and attitudes (UNESCO, 2022). In each dimension, mixed-level analysis was conducted, including qualitative descriptions of the whole-class learning performance and in-depth examination of individual students. The analysis started with reviewing the fieldnotes of classroom observations and transcripts of post-teaching debriefings to gauge overall classroom dynamics, student participation, and the effectiveness of the teaching strategies. Video-records of classroom interactions and transcripts of student interviews were retrieved to aid the analysis and clarify any ambiguities when necessary. This whole-class analysis provided a macroscopic view of how this approach was received and interacted with at a group level, highlighting general trends, common challenges, and shared successes (Candela et al., 2004).

Subsequently, the focus shifted to a micro-level analysis, where individual student cases were selected to represent a range of experiences within the classroom (Yin, 2011). These case studies involved a detailed examination of specific students' learning journeys, using data sources such as individual interviews, student work samples, and targeted observation notes. This approach allowed for a nuanced understanding of how different students engaged with and responded to this approach, uncovering varied learning pathways, personal challenges, and unique achievements. By integrating both whole-class and individual-level analyses, this mixed-level approach sought to capture the diversity of student experiences and provide a comprehensive, multifaceted understanding of the approach's impact (Good & Lavigne, 2017).

To enhance the reliability and credibility of qualitative analyses, triangulation was adopted throughout the analytical process (Baker & Green, 2007; Cater et al., 2014). The above analytical activities were co-conducted by two researchers who held regular meetings to compare and cross-validate their interpretations and findings. Triangulation across different data sources — including classroom observations, interviews, student work, and post-teaching debriefings — further strengthened the validity of the findings, allowing for cross-verification of data and insights gathered. After the analysis, the results were shared with teachers and other researchers for feedback and further validation. This step not only provided an additional layer of scrutiny but also fostered a collaborative environment where insights and interpretations could be challenged and refined.

5 Findings

The impact of the embodied constructionist approach on students' AI literacy is examined by comparing learning gains in the summer camp against their pre-instruction conceptions. The findings are presented with thick descriptions of student accounts and telling cases (Mitchell, 1984), for a detailed, depth understanding of the strengths and weaknesses of this approach.

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5.1 Students' pre-instruction conceptions of AI

To investigate students' baseline understandings of AI, their pre-instructional drawing pieces were examined by analyzing the depicted AI concepts, representations of AI functionalities, and any noted misconceptions. The analysis reveals that the conceptions of most students were superficial, fragmented, and naïve, including plenty of misinformation and erroneous beliefs about AI. Figure 3 shows three representative conceptions held by the students. Specifically, many students generally equated AI with technical products they had encountered in daily lives, such as the security camera, virtual reality goggles, and robot vacuum in Fig. 3A. Other examples included mobile phones, calculators, and quick-response (QR) codes. According to the student interview, they defined the key attributes of AI as being digital, electronic, or technological. Figure 3B shows another type of conception that AI was a duplicate of humans and AI shared the same attributes with humans. This conception reflects popular misinformation rooted in the historical, literal meaning of AI – to mimic human behaviors with machines (Saila, 1996). The third type of student conception, as shown in Fig. 3C, was humanoid robots. Behind the external form of humanoid robots, the students again assigned human attributes to AI and viewed AI as completely autonomous as humans. This collection of student conceptions shows the students, whose pre-instructional conceptions were influenced by the technological products and public voices surrounding them, had superficial and naïve understandings that were not scientifically compatible with AI concepts. Coupled with the limited understanding, students barely had concrete skills in AI programming and problem-solving.

Regarding the ethical dimension, the students could hardly verbalize any ethical principles in their drawing pieces and stimulus-recall interviews before the intervention.

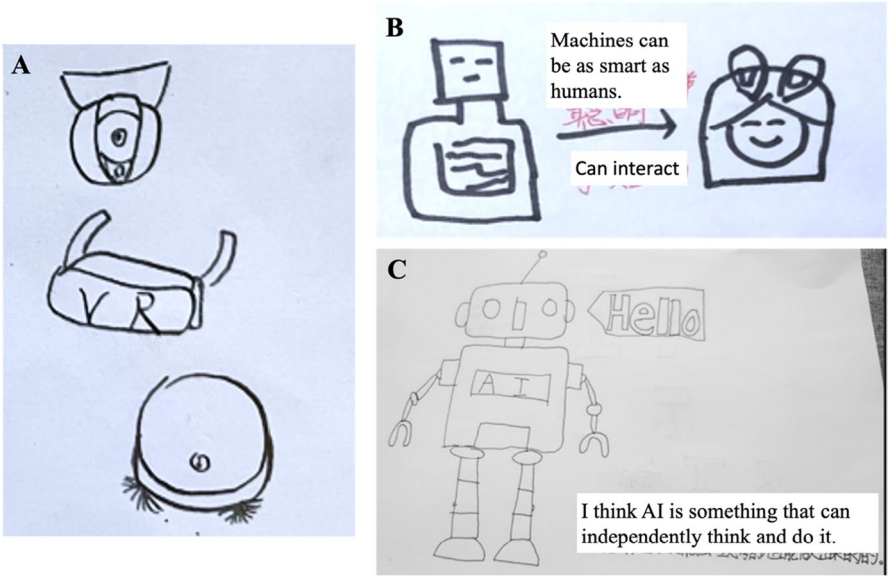


Fig. 3 Samples of students' drawing pieces

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When asked about "right" and "wrong" in developing and using AI, many students tended to assume that AI was inherently useful and helpful, as a student remarked, "AI is to help us do tasks and make our life more convenient." While this view was one-sided and overly optimistic, it to some extent reflects the ethical principle of beneficence that emphasizes the technology's potential to bring about positive outcomes and societal benefits. However, this optimism also shows a gap in the student's understanding of broader ethical considerations and potential risks in AI, such as issues related to privacy, bias, and accountability. Besides, some students were concerned that AI could be dangerous and harmful if controlled by evil powers. This concern was primarily caused by the science fiction cartoons or movies they had been exposed to, where AI is often portrayed as a tool of villainous characters or as a technology that spirals out of control. The observation shows a lack of ethical awareness and understanding among students, as they barely held any concrete, solid considerations regarding AI ethics.

5.2 Student development in AI literacy in the summer camp

Student development is examined in the conceptual, technical, and ethical dimensions, in alignment with the knowledge, skills, and values and attitudes of AI literacy.

5.2.1 Student understandings of AI concepts and processes

Student growth in AI knowledge was primary assessed via classroom observation (conducted by both the teacher and researchers), student assignments, and interviews, where data triangulation was performed to validate the assessment. The analysis shows that, after the intervention, the participating students demonstrated significant growth in their understanding of basic AI terms and concepts. Regarding AI concepts, students gained a general understanding of their meanings and ideas. This understanding was not a memorization of definition, but rather a conceptual grasp of fundamental attributes and implications. For example, in explaining what AI was, students were able to approach it as intelligent agents capable of sensing environments, reasoning, and taking respective actions. A student elaborated this understanding in comparing an electric fan and an air conditioner:

> *After we see something with our eyes, the first thing we do is use our brains to think and judge how to accomplish certain tasks. Then we can do that thing, or to avoid any trouble. This is adaptive behavior… The electric fan is not AI. You need to press the switch. It's automated because we have to manually press the switch to make it blow air, or manually adjust the intensity of the wind and the oscillation. It hasn't evolved into being adaptive yet. Then I saw the air conditioner with constant temperature control. It can automatically detect and adjust the temperature if it senses the temperature is too high or too low. I think it works in an adaptive way, it's intelligent. If it's autonomous, it adapts based on our actions. If they can always adapt on their own, then we don't need to do some things. You let it complete the task by itself.*

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In this interview, the student demonstrates an emerging understanding of the differences between simple automation and adaptive, intelligent behavior in AI systems. In analyzing how the electric fan and air condition functioned, she focused on the degree of human intervention and self-adaptiveness, as a key attribute of AI. Her grasp of the concept of adaptiveness was based on a comparative mapping between humans and AI. Her reasoning started with relating AI functions to familiar bodily actions and sensory experiences, such as seeing and adjusting behavior accordingly. Then she used her own experiences and observations as a framework to comprehend the working process of electric fans and air conditioners and identify which was intelligent. More importantly, it helps students to conceptualize AI not as a monolithic or abstract concept but as a spectrum of technologies with varying degrees of complexity and autonomy. In this way, students were able to grasp the fundamental attributes and meanings associated with AI concepts, rather than a memorization of definitions.

Moreover, students exhibited a deeper comprehension of the working mechanism and strategy of AI. Since ML was a major topic in this course, human learning had been intensively leveraged as a reference to explain processes and strategies of ML, such as observation of training data, extracting features and patterns, and object recognition. Using human learning as an analogy, students were able to grasp the idea that just as humans learn from experience, ML algorithms improve their performance by being exposed to and learning from data. This understanding was further reinforced in their constructionist project where they took photos of their own body gestures to train a ML model of gesture recognition. The constructionist project offered a hands-on experience of applying and testing their conceptual understandings at first hands.

In the process of training ML models, students also gained insights into the challenges and limitations of AI. They encountered issues such as overfitting, where the model performs well on training data but poorly on unseen, untypical data. This experience offered an opportunity to break down the human-AI analogy and trigger students to reflect on the strengths and limitations of AI. For example, a student commented about the difference between humans and AI:

> *Humans can learn quite quickly and can infer from one example to others. If you give them a picture, they can analyze and make judgments based on its characteristics, even if the picture is not very complete. However, machines are slower in this respect and can't make such inferences easily. They need a lot of pictures and extensive data. Their learning can sometimes be rigid. For instance, if you put something untypical in front of them, they might not be able to recognize it. But machines have one advantage: they have a better memory. As long as there are no destructive issues, the data in their database remains there and can be preserved for a long time. Humans, on the other hand, tend to forget easily and have limited memory capacity.*

This student's comment reflects a sophisticated understanding of the fundamental differences between human cognition and AI processing, as well as a critical understanding of the limitations of current AI technology. These insights imply the unique affordance of constructivist projects in enhancing student understandings. By engaging directly with AI systems and critically reflecting on their experiences, students could easily overcome the misconception that AI worked in an autonomous way like

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humans. In this way, students were more likely to gain a balanced understanding of AI's strengths and limitations and a realistic perspective on its potential role in society.

5.2.2 Student mastery of technical and problem-solving skills

The examination of student development in AI skills is based on their AI programming tasks in Scratch, with a focus on their technical skills and problem-solving skills. Compared with a general growth in conceptual understandings, they demonstrated a more nuanced and varied progression in skills, as the extent and nature of their skill development differed individually.

First, most students exhibited a growing technical familiarity and became more adept at using blocks to create algorithms and intelligent agents[1] and manipulating AI applications. In each workshop, students were given a tiered programming project in Scratch. In this project, students were first guided to work on a basic task of linear programming exercise, where they used blocks to actualize learned AI concepts and processes. Then for those advanced students who completed the basic task early and accurately, a more challenging task was provided. The advanced task required a higher level of problem-solving skills. Classroom observation showed that after the instruction, at least half of the students could independently finish the basic task and the remainder could finish with certain scaffolding. This outcome was considered promising, considering the novice status of the students in learning AI.

This growth in technical proficiency was primarily attributed to the hands-on programming exercises and the enhanced conceptual understandings. Especially, many students explained that, since embodied analogies had prepared their minds first, programming was less challenging for them, in comparison with their previous programming courses. A student commented:

> *Programming was not as difficult as I thought. I used to think that it was to follow teachers' steps to put together blocks as they did. (Now) I know why I need to sequence the blocks in a certain way, what actions they can do when these blocks are put together.*

This student account shows that the enhanced student understandings could demystify programming and illuminate the logical underpinnings of AI processes. In this way, programming was no longer a mere imitation of steps, but a meaningful exploration and application of understandings.

Along with the growing technical proficiency, were the emerging problem-solving skills as students sought to tackle problems with informed solutions in the advanced tasks. Overall, students showed considerable progress in formulating problems, developing solution paths, and choosing appropriate applications (if available), but there was a noticeable disparity in their ability to translate these ideas into practical implementations using the programming blocks. This gap can be best demonstrated with a telling case about the decision tree task. In this task, five types of familiar animals were provided, and students were requested to develop an

[1] In an AI programming task, the whole project (e.g., a door access control system) or a Spirit (an image that can be programmed in Scratch) is viewed as an intelligent agent.

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intelligent system for a zoo that could identify the animals. Figure 4 shows a student's hand-drawn decision tree (Fig. 4A) and Scratch project (Fig. 4B) in this task.

As shown in Fig. 4A, the manual decision tree was well-formed, showing a clear understanding of the logical process required to identify animals based on a set of binary questions. It represents the student's ability to formulate the problem and conceptualize a solution path – a critical step in AI problem-solving. However, he struggled with actualizing the conceptual solution into programming, leading to an incomplete Scratch task in Fig. 4B. The follow-up interview reveals his confusion in distinguishing "if…then" and "if…then…else" blocks. Since each node – a decision point – includes two different outcomes ("yes" and "no"), the "if…then…else" block is designed to handle exactly this type of binary decision-making process. But this student mistakenly chose the "if…then" block; omitting the "else" part, as the student did, means the program had no instructions on how to proceed when the condition failed, leading to a non-functioning decision-making process. Furthermore, a decision tree with multiple levels (identifying five animals) needed to nest "if…then…else" blocks within each other, which appeared too complicated for him. He got lost in tracking which conditions belonged to which branch of the decision tree and struggled with the cognitive load required to hold multiple pieces of information in mind at once in programming.

The decision-tree case contrasted with this student's performance in a subsequent ML task depicted in Fig. 5, which involved controlling a Spirit's movement in Scratch. Similar to his performance in the decision tree task, the student effectively formulated the problem into an image recognition model and identified a solution. However, unlike the prior task, he successfully implemented the solution using ML and other blocks in Scratch. A major difference between the two tasks was the programming complexity. The decision tree task required a deeper understanding of Scratch's programming constructs, specifically the conditional statements and their nesting, whereas the ML task offloaded some of the complexity to the ready-to-use ML blocks, allowing a focus on integrating and applying

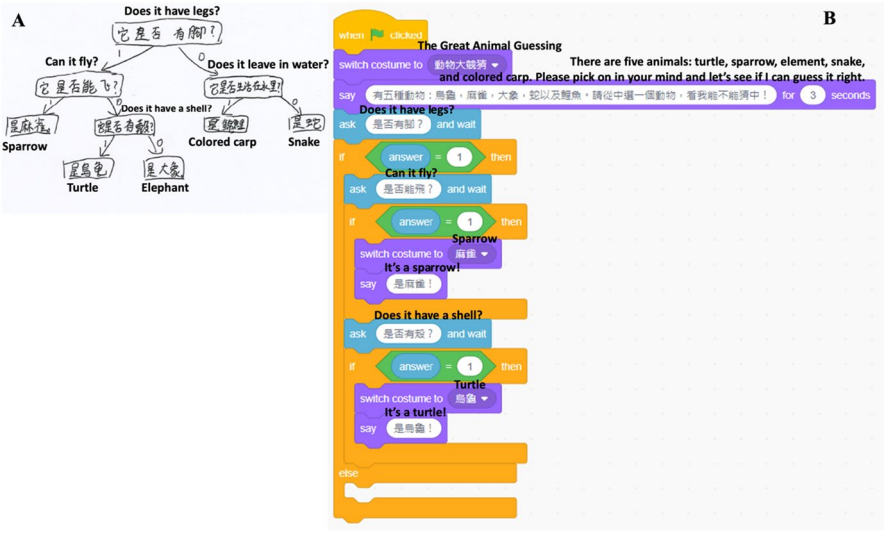


Fig. 4 Sample of a student's hand-drawn and programmed decision trees

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the ML model. This suggests that while the embodied constructionist approach prepared students for problem conceptualization and solution development, it may not fully prepare them for complex programming logic and technical implementation.

5.2.3 Student gains in ethical and critical understandings

Student learning in the ethical dimension was observed in their classroom participation, assignments, and interviews. The participating students showed a general increase in their ethical awareness and critical attitudes, as they started noticing, recognizing, and understanding the ethical implications of AI. As shown in the aforementioned analysis, the human-AI comparison and constructionist projects allowed students to reflect on the strengths and limitations of AI, leading to a more balanced, realistic understanding of AI. Such understandings could effectively resolve students' misconceptions and one-sided and overly optimistic views of AI, as revealed in the pre-instructional conceptions, and trigger their awareness and attention to the ethical challenges. For example, many students felt surprised to learn that AI could discriminate, as they used to think that machines were not as emotional or subjective as humans. A student explained:

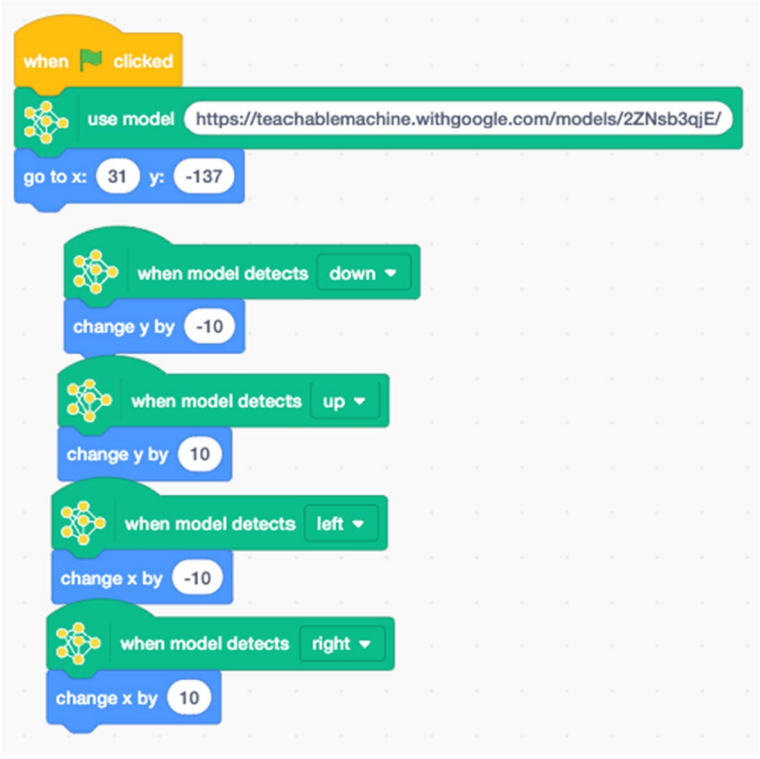


Fig. 5 Scratch project of a student in a machine learning (ML) task

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I know humans discriminate, but did not heard of machine discrimination before…Machine discrimination is heartless. (Why "heartless"?) Its decision-making is heartless, yes is yes, no is no, 1 or 0. Machines can only determine some features, like color and shape. But they don't know if this is a nice person. The decision is also influenced by the bias in training data, for example, prioritizing boys over girls. Such incorrect decisions may ruin people.

The student account reflects a nuanced comprehension beyond a general awareness or superficial understanding of ethical issues. In resituating discrimination from human contexts to machine decision-making, her characterization of AI decisions as "heartless" conveys an understanding of AI's binary logic – it operates on strict parameters set by its programming and data inputs – and preferences for quantifiable features like color and shape. Such understandings were built upon their conceptual understandings and technical proficiency with the working process of AI, which she pushed forward to scrutinize its implications for humans. This level of ethical understanding suggests that the student had internalized the complexities of AI ethics, moving beyond a cursory engagement with the topic to a more reflective and critical stance.

Beyond the ethical understandings, some students were able to apply such understandings to re-examine and reflect on their life experiences. For example, a student related his AI learning to gaming experiences:

Human decision-making can consider various factors, while what machines consider are objective, one-sided, or most important factors. For example, I've been playing a game recently called 'Second Galaxy.' Machines make decisions that are optimal for the immediate situations, unable to effectively handle emergencies. Humans can think long-term and provide effective solutions to sudden events. Once, I ran out of bullets during a mission. The game's system advised me to immediately give up the mission and return to the space station (with my warship). That was indeed the best solution at that moment, because my warship would be destroyed. But I didn't do that, as the mission offered huge rewards for which I had waited half a year, and it was not too difficult. Since my ship had high defense and enough energy, I had a war of attrition to defeat the enemy forces. The machine didn't consider all these, simply asking me to return to the space station. When we make simple decisions or need quick results, we can use machines. However, when dealing with emergencies and considering long-term impacts, we should think by ourselves, using machines as assistants.

This student's reflection demonstrates an ability to apply their understanding of AI ethics and decision-making processes to personal experiences. It showcases a higher level of cognitive development where abstract concepts are integrated into concrete contexts. Furthermore, his narrative shows an appreciation for the complementary roles that humans and machines can play. His learning had transcended theoretical knowledge to critically examine and reflect upon the interplay between AI capabilities and human judgment. This capacity for reflection is indicative of a deeper learning that can inform the responsible use and development of AI technologies in the future.

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6 Discussion and Conclusion

In this design-based research, we propose an embodied, constructionist approach to promote holistic AI literacy development in upper primary education. To further examine its actual impact on students, we implemented this approach in a local school and conducted a classroom ethnography. By qualitatively comparing student performance before and after the study, we found that overall, the participating students had general growth in their conceptual understanding, technical proficiency, ethical awareness, and critical attitudes; but they also struggled with complex programming and technical tasks. Interpretation and implications of the research findings are presented.

The overall positive impacts of the embodied constructionist approach might be attributed to three mechanisms. Firstly, the integration of unplugged (embodied analogies) and plugged (constructionist projects) activities created a rich, multi-dimensional learning environment for students at large. This integration not only catered to diverse learning styles but also bridged the gap between theoretical understanding and practical application (Ma et al., 2023). Secondly, the 'Understand–Make–Reflect' cycle embedded within this approach encouraged continuous learning, where students not only acquired knowledge but also applied it in creating AI-driven solutions, followed by reflecting on their experiences and outcomes (Duarte & Baranauskas, 2024). This cycle promoted critical thinking and problem-solving skills, as students were constantly challenged to assess their work. Lastly, the comparative mapping between human and AI processes has proven to be an effective tool in demystifying AI (Dai, 2024; Greenwald et al., 2024; Zhou et al., 2024). By drawing parallels between human cognition and AI algorithms, students could grasp complex concepts more intuitively. This comparative approach also aids in addressing common misconceptions and apprehensions about AI, fostering a more nuanced and balanced understanding of the technology and its capabilities.

This approach addresses the recent calls for holistic AI literacy and AI education programs in the K-12 sector (Touretzky et al., 2019; UNESCO, 2022). Forging holistic development of AI literacy, though widely accepted nowadays, remains a major challenge, especially among young learners (Casal-Otero et al., 2023). In this regard, the embodied constructionist approach offers a feasible solution to fill in the gap. Especially, compared to previous strategies of combining direct instruction with hands-on activities, this approach enables a cohesive journey with progressive, mutually reinforcing learning components (Yue et al., 2022). Moreover, the integration of ethical and critical reflections within this approach aligns with the growing emphasis on responsible AI education (Zhang et al., 2023). By contextualizing the constructionist projects (developing AI systems) in human-AI comparisons, the ethical and critical reflections were no longer mere "add-ons" to hands-on projects, but integral components of the foundational learning process (Ruppert et al., 2023). Regarding student struggles with complex programming tasks, AI literacy can be combined with computational thinking (CT) and programming education to enhance their learning experiences (Weng et al., 2024). By connecting AI principles with CT skills such as decomposition, pattern recognition, and algorithmic thinking, students can better translate conceptual understandings into practical programming experiences. This integration

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can promote deeper engagement and help students overcome barriers when tackling complex technical programs in AI-focused programming environments.

This research contributes to theoretical knowledge about embodied cognition and constructionism in the context of AI education. In previous studies, embodied cognition has been primary adopted in participant simulation activities or hands-on manipulation of AI tools (Druga et al., 2019; Yang et al., 2023). The embodied analogy in the present study, however, extends beyond the conventional adoption by constructing a conceptual mapping between bodily human experiences with corresponding AI processes. While this embodied mapping makes AI more accessible and relatable for younger students, it also fulfills curriculum objectives about exploring similarities and differences between humans and AI (Long & Magerko, 2020; Touretzky et al., 2019). In this regard, this study offers a domain-specific embodied approach for AI education. As for constructionism, this study expands its potential on supporting students' conceptual and ethical learning beyond the traditional focus on hands-on making (Valente et al., 2021). By embedding iterative cycles of understanding, making, and reflecting within the learning process, the research underscores the dynamic interplay between conceptual knowledge, practical skills, and ethical considerations, thereby expanding the scope of constructionism to address contemporary educational needs in AI (LaPrade & Lassiter, 2021).

There are limitations with the embodied constructionist approach and AI education research, pointing to directions for future studies. Specifically, while this approach enhanced students' AI literacy, it may fall short in preparing students for more abstract reasoning and technical skills in programming-based problem-solving environments. A major reason behind this limitation was that this approach, while focusing on students' sense-making and understandings about AI, might not have sufficiently addressed the higher-order thinking skills required for tackling complex programming challenges or for engaging in advanced problem-solving tasks (Grover & Pea, 2018). Besides, while this approach is more suitable for introducing basic AI topics, it may not provide the necessary depth to explain more intricate aspects of AI, such as algorithmic complexity and data structures (Dai, 2024). This gap suggests a need for integrating additional components into the curriculum that focus specifically on advanced computational thinking and programming skills. Such components could include problem-solving exercises that require more abstract thinking, direct instruction on advanced programming concepts, or projects that encourage students to apply AI in more complex and less guided contexts.

While this classroom ethnography offers thick descriptions and rich details about students' overall development, future studies can adopt quantitative methods or mixed methods to compare this approach with other conventional methods, such as a fully unplugged or plugged approach in teaching AI. Employing a quantitative approach would allow for the collection of measurable data, such as standardized test scores, to assess and compare the effectiveness of different teaching methods in enhancing students' AI literacy, programming skills, and conceptual understanding. Additionally, as this study has collected a rich dataset, more micro-level, interactional analyses can be conducted to delve into the moment-by-moment interactions among students and examine their sense-making in situ. Another area for future research is to explore how this approach could be adapted for diverse student populations, including those with different learning preferences, abilities, and socio-economic backgrounds. Research in this area could enhance the accessibility and inclusiveness of AI education, contributing to a more equitable education landscape.

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7 Appendix

Table 2 A summary of teaching content

#	AI concept	Constructionist project	Ethical reflection
1	• Understand the attributes of intelligent agents • Recognize AI applications in everyday life • Understand the basic function and types of sensors • Understand the role of sensors in AI	• Master the basics of using Scratch • Construct simple code scripts • Develop a computational agent capable of sensing virtual environments and avoiding obstacles in Scratch	• Become sensitive to the human-AI relationship • Develop awareness of beneficence and non-maleficence • Become sensitive to the interaction between AI, environments, and human society
2	• Understand the idea of binary numbers • Apply binary numbers to code daily information • Become sensitive to the processes of encoding and decoding	• Develop a system that uses binary numbers to represent features of surrounding objects and take respective actions in Scratch	• Understand the reductionist tendency in the data representation and its limitations and societal impacts
3	• Understand the idea of patterns and features • Understand the process of data labelling	• Develop a classifier to recognize fruits in the RAISE platform	• Develop awareness of human decisions and biases in feature selection
4	• Understand the idea and process of decision tree • Manually construct decision trees to represent the process of supervised learning	• Develop a decision tree-based system that can recognize animals based on the input of animal features in the RAISE platform	• Understand the impact of human bias in machine learning • Understand the strengths and limitations of machine learning
5	• Understand common features and patterns in written language • Understand the idea and application of large language model (LLM)	• Develop a system that can converse with an LLM-based chatbot in the RAISE platform	• Become sensitive to the potential applications of LLM in daily life • Understand the strength and limitation of LLM
6	• Understand the attributes and sources of big data • Understand the basic principle of data ethics • Understand the application of big data in machine learning	• Develop a system that can recognize facial expressions and use the recognition results to control the movement of agents in the RAISE platform	• Reflect on the personal footprint • Develop an awareness of cybersecurity • Critically evaluate unethical conduct related to data
7	• Understand basic recommendation approaches • Recognize recommender systems in daily life • Understand the idea and the types of rewards	• Develop a system that can capture various human gestures as positive/negative rewards and use the rewards to fine-tune the algorithms and decisions	• Critically evaluate the phenomenon of information cocoons and its impact on human lives
8	• Recognize the phenomenon of algorithm biases • Understand the source of such biases • Understand the impact of stakeholders in the development of AI systems	• Develop an agent that can adjust the weight of various features in student selection (based on the stakeholder's perspective) and use the agent to aid the decision-making	• Critically evaluate algorithm bias and its impact on users and society

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